carried out under pressure for deeper and uniform penetration of dye. This method is used for the manufacturing of denims, in which warp yarns are dyed with indigo and weft yarns are left undyed.

Advantage

- Economical than other yarn dyeing methods

Disadvantage

- May not produce high fashion fabric since warp yarn will be dyed in specific colour

Space Dyeing

In this method yarn is not completely dyed, but it is dyed at intervals. Space dyeing is done by two methods namely, Knit-de-knit method and OPI space -dye applicator method. In knit-de-knit method, the knitted fabric is dyed and de-knitted, which results in appearance of alternating dyed and un-dyed spaces. In OPI space-dye applicator method, the yarns are allowed to pass through space dye baths at very high speed which results in intermittent dyeing.

Advantages

- Produces brighter and deeper shades.
- Create interesting checks, stripes and plaids.

Disadvantages

- Costly and time consuming
- May not create designs to match current fashion

2.3.1.3 Dyeing at Fabric Stage

The process of dyeing woven or knitted fabric is called as **piece dyeing**. Piece dyeing methods include beam dyeing, beck or winch dyeing, jig dyeing, jet dyeing and pad dyeing.



Figure 2.5 Piece Dyeing

Advantages

- Uniform dyeing
- Inexpensive than fibre or yarn dyeing
- Versatile and flexible in accommodating changing fashion trends.
- Faster and easier

Disadvantage

- The ends of the fabric does not have some colour as the remaining fabric

Beam Dyeing

This process is similar to warp-beam dyeing of yarns. In this method, the woven or knitted fabrics are wound on perforated beams and immersed in dye solution. Dye is circulated through the fabric

Beck / Winch Dyeing

This method is done to dye the fabrics of longer length. The fabric is circulated through the dye bath in tensionless rope form. The ends of the fabric are tied together to form a loop. The fabric moves over a rail onto a winch reel which immerses it into the dye bath and draws the fabric up. The process is repeated till the fabric is dyed uniformly and to the desired colour shade. About 1000 m or 900 kgs of fabric can be dyed at a time. The dyed fabrics retain its original softness feel. This method is widely used for woolen woven and knitted fabrics.

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